

WTI CRUDE OIL FUTURES (CL)

Spread & Curve Structure — Fundamental Reference
Cushing, Shale, Pipelines, and the US Domestic Market

This document presents a fact-based reference on the fundamental drivers of NYMEX WTI Crude Oil futures (ticker: CL), with particular focus on the forces that shape the forward curve — from prompt spreads to the long end. WTI (West Texas Intermediate) is the primary US crude oil benchmark and the world's most actively traded crude futures contract by volume. It prices the physical crude oil market at Cushing, Oklahoma — a landlocked delivery hub whose inventory dynamics, pipeline connections, and refinery relationships are structurally different from seaborne benchmarks. These structural features make WTI spreads respond to a distinct set of physical fundamentals that this document covers in full.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Physical Delivery

WTI is a physically delivered contract. This is the most fundamental structural difference from ICE Brent, which is cash-settled. Physical delivery creates hard constraints on the prompt spread that are directly determined by pipeline capacity, tank availability, and logistics at the Cushing, Oklahoma hub.

Parameter	Detail
Exchange	CME Group — NYMEX (New York Mercantile Exchange)
Ticker	CL (front month); also traded as WTI on ICE Futures US (ticker: T)
Contract size	1,000 barrels (42,000 US gallons)
Quotation	USD per barrel, minimum tick \$0.01 (= \$10 per contract)
Contract months	All months out to approximately 9–10 years forward; active liquidity to ~36 months
Settlement type	Physical delivery at Cushing, Oklahoma — pipeline or storage transfer
Delivery grade	Domestic sweet crude (40.0–44.9° API, max 0.42% sulphur); WTI, Low Sulphur Sweet, New Mexican Sweet, North Texas Sweet, Oklahoma Sweet, South Texas Sweet eligible. Foreign crudes permitted at differentials.
Delivery location	Cushing, Oklahoma — the "Pipeline Crossroads of the World." Connected to ~35 pipeline systems and 90+ million barrels of storage capacity.
Trading hours	CME Globex: Sunday–Friday 18:00–17:00 ET (23-hour session); pit session 09:00–14:30 ET (historical, now largely electronic)
Expiry / last trade	3rd business day prior to the 25th calendar day of the month preceding the delivery month
Open interest profile	Largest concentration in months 1–6; meaningful to 24 months; activity tapers significantly beyond 36 months

1.1 Why Physical Delivery Matters for Spreads

Because WTI is physically delivered at Cushing, the relationship between physical inventory levels and pipeline infrastructure determines the hard limits on the forward curve structure in a way that cash-settled contracts do not face:

- When Cushing storage is full or approaching capacity, sellers of the front contract cannot easily find storage for physical delivery. This forces prices down — in extreme cases, negative (April 2020). The prompt spread goes into steep contango as sellers pay a premium to push barrels into future months.
- When Cushing storage is very low (tight inventory), buyers of the front contract face difficulty sourcing physical barrels for delivery. The front contract is bid up, creating backwardation. The spread between C1 and C2 widens in favour of the front.
- The 25th-of-month expiry rule creates a specific calendar dynamic: positions must be closed or rolled well before expiry, and the approach of expiry creates mechanical roll pressure that is predictable in timing if not magnitude.

1.2 Key Crude Grades Eligible for Delivery

Multiple domestic and some foreign crude streams are deliverable against WTI. The differential between each eligible grade and WTI par determines which barrels are most economic to deliver:

- **WTI Cushing** — the par delivery grade. Produced in the Permian Basin (West Texas/New Mexico) and other US plays, transported to Cushing via pipeline.

- **WTI Midland** — a Permian Basin grade that has grown substantially with the shale boom. Trades at a differential to WTI Cushing that reflects pipeline capacity, local supply conditions, and export competition. Since 2023, WTI Midland is included in the Platts Dated Brent assessment, linking its differential directly to the Brent-WTI relationship.
- **Light Louisiana Sweet (LLS)** — Gulf Coast sweet crude. Historically trades at a premium to WTI Cushing due to its proximity to Gulf Coast refinery demand and export terminals. LLS-WTI spread reflects the economics of moving barrels from Cushing to the Gulf Coast.
- **Mars (medium-sour)** — Gulf of Mexico sour crude. Not deliverable against WTI but important as a benchmark for US sour crude pricing and a reference for refinery yield economics.

2. Cushing, Oklahoma — The Delivery Hub

Cushing is the physical heart of the WTI market. Its inventory levels, pipeline connections, and storage utilisation are the most important single physical variable for WTI prompt spread behaviour. Understanding Cushing as a system — not just a number — is the foundation of WTI spread analysis.

2.1 Cushing Storage Infrastructure

Parameter	Detail	Spread Implication
Total working storage capacity	~76-90 million barrels (varies with infrastructure investment)	Structural upper bound on Cushing inventory; approaching capacity = contango pressure
Operational capacity (practical limit)	~80% of total = typically ~60-70 mb	Above ~60 mb historically creates strong contango pressure in C1-C2
Pipeline connections	~35 pipeline systems: Seaway, Keystone, Midland-to-Echo, Ozark, Longhorn, Centurion, White Cliffs, STACK, SCOOP pipelines	Pipeline reversals and capacity expansions change the structural flow of crude to/from Cushing
Export access	No direct pipeline to deep-water export terminal. Cushing crude must move to Gulf Coast (Seaway, Longhorn etc.) before export.	Pipeline tariff to Gulf Coast is embedded in the WTI-LLS differential
Key storage operators	Plains All American, Magellan, Enbridge, Vitol, Enterprise, Sunoco — tank space leased commercially	Tank lease rates influence the economics of contango storage plays

2.2 The Cushing Inventory Cycle

Cushing inventories are not static — they move in response to upstream production changes, pipeline flows, refinery demand, and export volumes. Understanding the direction and pace of inventory change is more important than the absolute level for near-term spread positioning:

- **Inventory builds:** more crude arriving at Cushing than leaving. Occurs when: upstream production rises faster than pipeline throughput, refinery runs fall (maintenance, margin collapse), or Midcontinent refinery demand is weak. Sustained builds push the curve toward contango and weaken the C1-C2 spread.
- **Inventory draws:** more crude leaving Cushing than arriving. Occurs when: Midcontinent refineries run at high utilisation, Gulf Coast export demand pulls barrels south via Seaway/Longhorn, or PADD 2 (Midwest) refinery throughput is elevated. Sustained draws create backwardation pressure in C1-C2.
- **Seasonal pattern:** Cushing inventories typically build during Q1-Q2 (refinery maintenance season, lower runs) and draw during Q3-Q4 (high refinery utilisation, peak driving/heating demand). This pattern is not guaranteed — production changes, pipeline reversals, and export volumes can override the seasonal tendency in any year.
- **The 5-year average comparison:** the EIA reports Cushing stocks alongside the 5-year average range. Stocks at the top of the 5-year range signal potential contango; at the bottom, backwardation. The direction of deviation matters as much as the level.

2.3 Pipeline Infrastructure — The Physical Plumbing of WTI

WTI spread behaviour cannot be understood without knowledge of the major pipelines that move crude to, from, and through Cushing:

- **Seaway Pipeline** (Cushing to Freeport, TX — Enterprise/Enbridge): reversed in 2012 from northbound to southbound. This single event structurally changed the WTI-Brent relationship, allowing Cushing surplus to flow to Gulf Coast refineries and export terminals. Seaway capacity is approximately 850,000 b/d. When Seaway runs at capacity, it limits the degree of Cushing builds.
- **Keystone Pipeline** (Alberta to Cushing and Gulf Coast — TC Energy): brings Canadian heavy sour crude (Western Canadian Select, WCS) into the Cushing/Midwest system. Canadian crude volumes arriving at Cushing contribute to inventory builds when Midcontinent refinery demand is insufficient to absorb both domestic and Canadian supply.
- **Longhorn Pipeline** (Permian Basin to Houston — Magellan): moves Permian crude directly to Gulf Coast, bypassing Cushing. Increased Permian-to-Gulf-Coast capacity reduces the fraction of Permian growth that reaches Cushing, muting the inventory impact of shale growth.
- **Midland-to-Echo (M2E) and other Permian Basin pipelines**: a network of pipelines moving crude from the Permian Basin to Gulf Coast export terminals (Corpus Christi, Houston Ship Channel). Capacity expansions in this system (e.g. Cactus II, EPIC, Gray Oak) directly affect the WTI Midland differential and Cushing inventory dynamics.
- **Flanagan South / Ozark** (Enbridge): moves Canadian crude from Illinois to Cushing and to Gulf Coast refineries. Canadian pipeline capacity changes (Enbridge Line 3 expansion) alter the volume of Canadian crude reaching the US Midcontinent.

Pipeline capacity is not constant — it is expanded, reversed, and constrained by regulatory decisions, legal challenges, and capital investment cycles. Major pipeline events (reversals, expansions, outages) are structural changes that can reprice the WTI curve for months.

3. US Shale and Tight Oil Production

US tight oil production — primarily from the Permian Basin, Eagle Ford, and Bakken — is the most important non-OPEC supply variable globally and the defining structural feature of the WTI market since 2010. Unlike conventional production, shale responds to price signals with a shorter lag, creating a dynamic supply-response mechanism that fundamentally affects the WTI forward curve.

3.1 Major US Producing Basins

Basin	Location	Production (approx.)	Crude Quality	Route to Market	Key Data Source
Permian Basin	West Texas / SE New Mexico	5.5–6.5 mb/d	38–42° API, sweet	Pipeline to Cushing OR direct to Gulf Coast (Midland-to-Echo, EPIC, Gray Oak)	EIA DPR, Enverus, operator guidance
Eagle Ford	South Texas	1.0–1.2 mb/d	40–50° API, sweet	Pipeline to Gulf Coast refineries and Corpus Christi exports	EIA DPR, Texas RRC
Bakken	North Dakota / Montana	1.1–1.3 mb/d	41–43° API, sweet	Pipeline (Dakota Access, ETCO) to Cushing and Patoka, IL; rail to East Coast	EIA DPR, NDIC
DJ Basin	Colorado / Wyoming	0.4–0.5 mb/d	40–42° API, sweet	Pipeline to Cushing and Colorado refineries	EIA DPR, Colorado OGCC
Anadarko / SCOOP/STACK	Oklahoma	0.3–0.4 mb/d	Mixed	Local Cushing delivery, well-positioned for hub	EIA DPR, Oklahoma CC
Haynesville / Other	Louisiana / misc.	Misc.	Varies	Varies	EIA state-level data

3.2 The Shale Production Response Mechanism

Tight oil production has a shorter price-response cycle than conventional oil but longer than often assumed. The key sequence:

- **Drilling decision to first production: 3–6 months** for a typical shale well. Once a well is spud, production will follow in 3–6 months regardless of subsequent price moves. This creates a near-term supply "pipeline" that is largely predetermined.
- **DUC (Drilled but Uncompleted) wells** — wells that have been drilled but not yet hydraulically fractured. The DUC count acts as a buffer: when prices rise, producers can complete DUCs quickly (2–4 months) without the full 3–6 month drilling cycle. A large DUC inventory allows faster production response to price signals. The EIA reports DUC counts monthly.
- **Rig count as leading indicator:** the Baker Hughes weekly rig count (active drilling rigs by basin and type) is the primary leading indicator of shale production 3–6 months forward. Rising rig count → rising production in subsequent quarters. Rig count responds to price with a 4–8 week lag.
- **Completion efficiency improvements:** lateral lengths, proppant volumes, and stage counts have increased materially over time, meaning the same rig count generates more production today than 5 years ago. Well productivity data (EIA Drilling Productivity Report) tracks this evolution per basin.
- **Operator breakeven costs by basin:** full-cycle breakeven costs (including land, drilling, completion, and transport) vary by basin and operator. Permian Basin Tier 1 acreage breakevens are estimated at \$35–45/bbl; marginal Bakken and DJ Basin wells are higher at \$50–60/bbl. These thresholds define where production growth slows or contracts in a lower-price environment.

3.3 The EIA Drilling Productivity Report (DPR)

The EIA's monthly Drilling Productivity Report is the most important single document for US shale supply analysis. It reports:

- New-well oil production per rig by basin — a direct measure of well productivity trends.
- Legacy decline rate — the natural production decline from existing wells per basin. US shale wells decline at 60–80% in the first year, creating a large "treadmill" of new drilling required just to maintain flat output.
- Net production change = new-well production – legacy decline. When new drilling cannot offset the legacy decline, total production falls even without a price collapse.
- Total production estimate by basin, current month and 1-month forecast.

The legacy decline rate is the most underappreciated feature of shale supply. A US shale production base of 13 mb/d declining at 4–5% per month (industry-wide average) requires roughly 600,000–650,000 b/d of new production each month simply to stay flat. This maintenance requirement is permanently bullish for the mid-curve when rig counts fall.

3.4 WTI Exports — The New Variable Since 2016

The US lifted its crude oil export ban in December 2015. Since then, US crude exports have grown from near zero to approximately 3.5–5.0 mb/d (2023–24). This has structurally changed the WTI market:

- WTI is no longer purely a domestic benchmark — it now competes globally with Brent, ESPO, Murban, and other seaborne grades. US export economics determine whether WTI Midland moves to Asia, Europe, or Latin America, and at what price relative to Brent.
- The primary US export grade is WTI Midland (Permian Basin). Houston Ship Channel and Corpus Christi are the primary export terminals. The Midland-to-Houston differential reflects pipeline bottlenecks and export terminal capacity — a recurring source of localised spread opportunity.
- Export demand provides a price floor for WTI: when WTI is cheap enough relative to Brent (WTI Midland discount to Dated Brent wide enough to cover freight and terminal costs), export demand increases, reducing Cushing/Gulf Coast inventories and supporting WTI.
- Weekly EIA crude export data (published in the Wednesday report) is now a key variable alongside domestic inventory data.

4. Demand Fundamentals

WTI pricing reflects US domestic crude demand filtered through the Cushing delivery point. However, because WTI is now an export benchmark, global demand also affects its price. The following demand variables are documented as price drivers.

4.1 US Refinery Demand — PADD Regions

The EIA divides the US into five Petroleum Administration for Defense Districts (PADDs). Refinery crude demand in each PADD affects the flow of crude to, from, and through Cushing:

PADD	Region	Refining Capacity	Crude Source	WTI Curve Impact
PADD 1	East Coast	~1.1 mb/d	Imported seaborne crude; some domestic via rail/pipeline	Minimal direct; monitors WTI-Brent arb for import economics
PADD 2	Midwest (Cushing area)	~4.0 mb/d	Primary WTI Cushing consumer; Canadian heavy crude via pipeline	Most direct impact — PADD 2 utilisation rate is a primary Cushing draw signal
PADD 3	Gulf Coast	~10.3 mb/d	Domestic WTI (via Seaway/Longhorn), imported heavy crude, Gulf of Mexico	Pulls barrels from Cushing southward; export terminal hub for WTI
PADD 4	Rocky Mountain	~0.6 mb/d	Local Rockies crude, some Bakken	Limited Cushing impact; local market dynamics
PADD 5	West Coast / Alaska	~2.9 mb/d	Alaskan North Slope, California, imports; isolated from Cushing	Minimal direct; separate pricing from Cushing hub

PADD 2 (Midwest) refinery utilisation is the single most important domestic demand variable for Cushing inventories. High PADD 2 utilisation draws Cushing stocks; low utilisation (maintenance, margin compression) allows stocks to build. EIA weekly data breaks out crude runs and utilisation by PADD.

4.2 US Gasoline and Distillate Demand

- **Gasoline demand:** approximately 8.5–9.0 mb/d in the US. The EIA reports weekly implied gasoline demand (product supplied). Seasonal peak in summer (Memorial Day to Labor Day). Gasoline demand has shown structural softening in recent years due to improved vehicle fuel efficiency and growing EV penetration, though the pace of change has been gradual.
- **Distillate (diesel/heating oil) demand:** approximately 3.8–4.2 mb/d. Driven by trucking, rail freight, agriculture, and industrial activity. Sensitive to economic cycle (GDP, freight volumes). Seasonal peak in Q4 (heating oil, agricultural harvest transport).
- **Jet fuel demand:** approximately 1.5–1.7 mb/d. Seasonal peak in Q3 (summer travel). Recovered fully from COVID-related trough by 2023.
- **Product stock levels:** EIA reports weekly gasoline, distillate, and jet fuel inventories separately from crude. Low product stocks (especially distillate) can support crude runs and pull Cushing inventories down even when crude stocks are ample.

4.3 Export Demand

- **Crude exports** (~3.5–5.0 mb/d): primarily WTI Midland to Europe, Asia, Latin America. Export demand responds to the WTI-Brent differential and freight economics. Weekly EIA export data is the most timely signal.
- **Product exports** (~6.0–7.0 mb/d): the US is a large net exporter of refined products (gasoline, distillate, LPG). Strong product exports pull crude demand (refineries run more crude to produce exportable products).
- **LNG-related diesel demand**: US LNG liquefaction facilities use diesel generators and affect local Gulf Coast distillate demand — a minor but occasionally visible signal.

5. Storage Economics and the WTI Carry Structure

Because WTI is physically delivered at Cushing, the storage economics at that specific location determine the carry structure of the front of the curve. This is more localised and more directly observable than for cash-settled benchmarks.

5.1 The Full-Carry Contango for WTI

The cost of storing WTI at Cushing and selling it forward consists of:

Cost Component	Approximate Range	Notes
Tank storage lease (Cushing)	\$0.18–0.35/bbl/month	Varies by tank operator, lease term, capacity availability. Rises sharply when storage fills.
Financing cost (interest)	Fed Funds rate-dependent	At 5% Fed Funds rate: ~\$0.35–0.40/bbl/month on a \$80/bbl barrel. Significant at high rates.
Insurance	\$0.01–0.03/bbl/month	Minor component.
Pipeline/throughput fees	\$0.02–0.05/bbl/month	For injecting and withdrawing from storage.
Quality loss / evaporation	Minimal for crude	Less relevant than for refined products.
Total full-carry cost	~\$0.55–0.80/bbl/month	At high interest rates and full tanks; lower when financing cheap and storage available.

- When the C1–C2 spread is in contango exceeding full carry, a cash-and-carry storage arbitrage becomes available: buy physical at Cushing, store, sell the deferred contract forward. This trade is executed by physical traders and financial institutions with tank access. The trade's existence places an upper limit on the contango — once the contango is wide enough to cover full carry, storage arbitrage compresses it.
- When Cushing tanks are near capacity (physical constraint), storage lease rates spike, raising the full-carry cost. This occurred during April–May 2020, when full-carry cost reached \$1.00+/bbl/month and the May 2020 WTI contract traded to –\$37.63/bbl on its expiry day — a historically unprecedented event caused by physical storage constraints at Cushing meeting COVID demand collapse simultaneously.
- The 2020 negative price event was specific to the physical delivery mechanics of WTI. ICE Brent, being cash-settled, did not go negative because no physical delivery at a landlocked hub was required.

5.2 EIA Weekly Inventory Data — The Primary WTI Signal

The EIA Weekly Petroleum Status Report, released every Wednesday at 10:30 ET, is the most important scheduled data event for the WTI curve. It covers the week ending the prior Friday and includes:

- **Crude oil stocks (total US):** headline number. Includes Cushing, SPR, and all PADD regions combined.
- **Cushing, OK crude stocks:** the most directly relevant number for WTI prompt spread behaviour. Reported separately from total US stocks.
- **Crude oil imports (total and by PADD):** shows how much foreign crude is entering the US system. High imports + weak refinery runs = builds.
- **Crude oil exports:** shows how much domestic crude is leaving. Strong exports = draws on US/Gulf Coast stocks.
- **Refinery inputs and utilisation rate (by PADD):** the single most important demand variable for crude. High runs + flat imports + rising exports = Cushing draws.

- **Gasoline, distillate, jet fuel stocks:** product inventories provide context for refinery incentives and end-user demand.
- **Net production estimate:** EIA's weekly production estimate (adjusted monthly in the more accurate 914 survey). Directionally useful but revised materially.

The market typically prices in consensus EIA expectations in advance (from the Tuesday API report and broker surveys). The spread move on the EIA release is proportional to the surprise versus consensus, not the absolute inventory level.

6. OPEC+, Geopolitics, and External Supply Drivers

While WTI is a domestic US benchmark anchored at Cushing, it does not trade in isolation from global crude markets. OPEC+ production policy and geopolitical events affect WTI through their impact on global supply/demand balance, import economics, and the WTI-Brent price relationship.

6.1 OPEC+ Policy Impact on WTI

- **OPEC+ production cuts:** reduce global supply, which tightens the global balance and reduces the availability of competing crude grades for US Gulf Coast refineries. This supports WTI indirectly by reducing import competition and widening the Brent premium (incentivising WTI exports to fill the gap in international markets).
- **OPEC+ production increases:** increase global supply, typically compressing the Brent-WTI spread as global crude becomes more abundant and US export demand for WTI softens.
- **Saudi OSP (Official Selling Price):** Saudi Aramco publishes monthly OSPs for its crude delivered to different regions. OSP changes signal Saudi pricing intentions and directly affect the competitiveness of Arab Light vs WTI for Asian and European buyers.
- **UAE, Iraq, Kuwait OSPs:** similar function to Saudi OSPs. Gulf producer pricing decisions together determine whether Middle Eastern crude is competitive with WTI for specific refinery markets.

6.2 Canadian Heavy Crude — WCS and the Cushing Impact

Canada is the largest single source of US crude imports, supplying approximately 3.8–4.3 mb/d into the US, primarily Western Canadian Select (WCS) — a heavy, sour blend from Alberta oil sands:

- **WCS-WTI differential:** WCS trades at a persistent discount to WTI reflecting its heavier, sourer quality and landlocked transport costs. The differential fluctuates with pipeline capacity, refinery configuration (heavy crude upgrading capacity), and Canadian production levels. Wide WCS-WTI discount benefits US Midcontinent heavy crude refiners (PADD 2) who process Canadian crude at high margins.
- **Keystone Pipeline (TC Energy):** the primary artery for Canadian crude into Cushing. Outages on Keystone (spills, operational shutdowns) create sudden supply disruptions at Cushing — reducing inventory builds and supporting the prompt WTI spread temporarily.
- **Trans Mountain Pipeline expansion (TMX):** completed 2024. Provides Canadian producers with Pacific coast access to Asian markets, reducing their dependence on US pipelines and potentially affecting the volume of Canadian crude directed to Cushing.
- **Rail transport:** when pipeline capacity is insufficient, Canadian crude moves by rail — at higher cost. Rail economics affect the WCS-WTI differential when pipelines are full.

6.3 Gulf of Mexico Production

Deepwater Gulf of Mexico (GoM) production is an important source of US crude that predominantly flows to Gulf Coast refineries rather than through Cushing:

- GoM production is approximately 1.8–2.0 mb/d, primarily medium-sour grades (Mars, Poseidon, Southern Green Canyon). These grades flow directly to PADD 3 refineries and compete with imported medium-sour crude (Iraqi Basra, Colombian Vasconia).
- Hurricane season (June–November) creates annual disruption risk to GoM production and Gulf Coast refinery operations. A major hurricane track through GoM oil infrastructure can remove 300,000–800,000 b/d temporarily, tightening the Gulf Coast physical market and supporting WTI prompt spreads.
- GoM production has longer development cycles than shale (3–7 years from discovery to first production), making it a back-curve supply signal rather than a prompt indicator.

7. Macro, Financial, and Cross-Market Drivers

7.1 The WTI-Brent Spread

The WTI-Brent differential (Brent minus WTI) is one of the most closely watched relationships in crude markets and directly reflects the relative supply/demand dynamics of the two benchmarks:

- **Structural factors:** WTI was historically at a premium to Brent as a higher-quality light sweet crude. Post-2010, the US shale boom flooded Cushing with domestic crude while the export ban prevented outflows, turning WTI into a persistent discount to Brent (2011–2015). The export ban lift (2015) and subsequent pipeline expansions normalised the relationship.
- **Current dynamics:** with WTI Midland now in the Dated Brent basket, the two benchmarks are more tightly linked structurally. The WTI Cushing vs ICE Brent spread reflects primarily: Cushing-specific inventory dynamics, Gulf Coast pipeline economics, and the freight cost of shipping WTI from Gulf Coast to European delivery.
- **Brent premium widens:** typically signals Cushing-specific oversupply (pipeline constraint, refinery outage) or geopolitical risk premium on Brent (Middle East tension). Wide Brent premium incentivises US exports.
- **WTI premium narrows or reverses:** signals either global demand weakness (reducing the Brent premium) or Cushing draws bringing WTI into line with seaborne crude values.

7.2 WTI Midland vs WTI Cushing (MEH Spread)

The Midland-to-Echo Houston (MEH) spread, or WTI Midland vs Cushing, reflects the cost of moving crude from the Permian Basin to the Gulf Coast export hub. This differential is highly sensitive to:

- **Pipeline capacity utilisation:** when Permian Basin pipelines run full, the Midland discount to Cushing/Houston widens as trapped barrels cannot reach market. This occurred acutely in 2018 when Permian production outpaced pipeline buildout, pushing WTI Midland to a \$15–18/bbl discount to Cushing.
- **New pipeline startups:** when new Permian-to-Gulf Coast capacity comes online, the differential compresses rapidly (as seen with Cactus II, EPIC, Gray Oak, and Permian Highway Pipeline).
- **Export terminal capacity:** physical constraints at Corpus Christi or Houston Ship Channel can trap barrels at the Gulf Coast, widening the Midland differential even if pipeline capacity is adequate.

7.3 US Dollar Impact

The WTI-USD relationship mirrors the Brent-USD relationship: a stronger USD typically exerts downward pressure on USD-denominated oil prices by reducing the purchasing power of non-USD buyers. However, for WTI specifically, the USD effect is moderated by the domestic nature of the market:

- For prompt WTI spreads (C1–C2), the USD effect is secondary to Cushing-specific supply/demand variables — a strong USD does not directly change Cushing inventory levels.
- For mid-to-back curve WTI (C6–C24), the USD effect is more relevant because it affects global demand expectations and OPEC+ production incentives (OPEC members' fiscal breakeven prices are in USD, so a weak USD effectively reduces their purchasing power, creating incentives to maintain or increase production volumes to compensate).

7.4 Interest Rates and Storage Economics

As detailed in Section 5, interest rates directly affect the full-carry cost of storing crude at Cushing. Rising interest rates increase the financing cost component of carry, which has two observable effects:

- The contango required to incentivise storage is higher, meaning the same physical surplus generates a steeper contango in a high-rate environment than in a low-rate environment.

- Conversely, backwardation can persist at a shallower level before it becomes economic to draw stocks — because the cost of holding crude is high, holders draw inventories at a lower backwardation threshold.

7.5 Speculative Positioning — COT Data

NYMEX WTI Commitments of Traders data is published weekly by the CFTC (Fridays, reflecting Tuesday close positions):

- **Managed money net position:** hedge funds and CTAs. Extreme net long: crowding risk, elevated reversal probability. Extreme net short: short-covering rally risk. These are probabilistic signals, not deterministic triggers.
- **Producer/merchant hedging:** US oil producers hedge forward production in WTI futures. Heavy producer hedging in deferred months suppresses back-curve prices and compresses backwardation. When producers are under-hedged (low price period, balance sheet stress), the absence of selling pressure allows back spreads to firm.
- **Swap dealer position:** reflects OTC hedging demand from producers and financial products. Large swap dealer short positions typically correspond to producer hedging activity.
- **Changes in open interest:** a sharp increase in open interest at a specific contract month often precedes a catalyst event at that horizon (e.g. OPEC meeting, scheduled refinery maintenance).

8. WTI Forward Curve Structure — Zones and Drivers

Zone	Contracts	Primary Driver	Secondary Driver	Key Signal	Typical Volatility
Prompt	C1-C2	Cushing inventory level and direction of change, PADD 2 refinery runs	Pipeline inflows/outflows, import economics	EIA Wednesday Cushing stocks; API Tuesday	Highest
Near-term	C2-C6	US production growth/decline, refinery utilisation trend, export pace	OPEC+ compliance, WTI-Brent spread	EIA STEO, DPR, rig count, PADD 2/3 runs	High
Mid-curve	C6-C12	US shale supply response to price, Canadian pipeline capacity, OPEC+ policy	Global demand cycle, USD, China imports	EIA STEO, IEA OMR, Enverus/Rystad supply data	Moderate
Back curve	C12-C36	Long-cycle capex decisions, GoM deepwater project startups, demand outlook	Energy transition pace, OPEC+ long strategy	Operator capex guidance, IEA WEO	Lower
Long-dated	C36+	Energy transition expectations, long-run demand trajectory	OPEC+ strategy, peak demand timing assumptions	IEA/EIA long-term outlooks, NDCs, policy	Lowest

8.1 Prompt Spread Mechanics

- The WTI C1-C2 spread is the most mechanically driven spread in crude markets: it moves predictably with Cushing inventory direction. Building stocks → contango pressure. Drawing stocks → backwardation pressure. The relationship is not perfect, but it is more reliable than any single factor in other commodity markets.
- The approach of the expiry date (3rd business day before the 25th) creates mechanical roll pressure on the front contract. Non-commercial participants must roll or close. Physical participants take delivery or negotiate EFP transactions (Exchange of Futures for Physical). This roll pressure is most visible in the final week before expiry.
- The volume-weighted roll from the front month to the second month is observable in open interest data. A rapid collapse of front-month open interest vs growing C2 open interest signals approaching expiry mechanics.

8.2 The WTI Butterfly

- WTI butterflies capture the curvature at specific points. Common fly configurations: 1-2-3 month (prompt fly), 6-12-18 month (mid-curve fly), and custom configurations around known events (OPEC meetings, refinery maintenance windows).
- Seasonal refinery maintenance creates predictable quarterly flies: Q1 maintenance (January-March) means PADD 2/3 runs fall, which builds Cushing stocks — negative for the prompt fly if both adjacent months are tighter. Q2 run ramp-up creates the mirror effect.
- Hurricane season (August-October) creates option-like risk on Q3/Q4 butterflies: if a major hurricane materialises, Gulf Coast production and refinery runs are simultaneously disrupted, creating complex, sometimes offsetting effects on the prompt curve.

8.3 Seasonal Patterns

Period	Key Seasonal Driver	Typical Cushing Effect	Typical Curve Effect	Reliability
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Jan-Mar	Refinery maintenance; post-holiday demand trough; cold weather heating demand offsets	Builds common — runs fall, production steady	Contango pressure in C1-C2	Moderate
Apr-May	Refinery turnarounds complete; US driving season begins; runs ramp	Draws resume — runs increase, pulling Cushing stocks	Backwardation builds or contango compresses	Moderate-High
Jun-Aug	Peak summer driving demand; hurricane season risk; high refinery utilisation	Draws typical — peak crude demand season	Backwardation at seasonal peak; prompt strong	Moderate
Sep-Oct	End of driving season; autumn refinery maintenance; hurricane risk extends	Variable — transition period with offsetting factors	Spread softens briefly during maintenance	Low-Moderate
Nov-Dec	Heating demand; year-end positioning; OPEC meeting often in this window	Variable — OPEC outcome can dominate	OPEC-dependent; often volatile year-end positioning	Low (event-driven)

WTI seasonal patterns are moderate in reliability. The Cushing inventory cycle creates real seasonal tendencies, but US production changes, pipeline developments, export volume swings, and OPEC+ actions all override the seasonal baseline frequently. Seasonal patterns provide a probability framework, not a trading signal.

9. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Release Timing	Primary Curve Impact
Weekly Petroleum Status Report	US EIA	Weekly	Wed 10:30 ET	Prompt C1-C3; Cushing stocks the key number
API Weekly Statistical Bulletin	API	Weekly	Tue ~16:30 ET	C1-C2 consensus-setter; surprise vs EIA matters
Short-Term Energy Outlook (STEO)	US EIA	Monthly	2nd Tue of month	C3-C12; US supply/demand balance projection
Drilling Productivity Report (DPR)	US EIA	Monthly	~2nd Mon of month	C3-C12; shale production and DUC count by basin
914 Production Survey	US EIA	Monthly	~1 month lag	C3-C6; most accurate US production estimate
Baker Hughes Rig Count	Baker Hughes	Weekly	Fri ~13:00 ET	C3-C12; leading indicator for shale production
IEA Oil Market Report (OMR)	IEA	Monthly	~2nd week of month	C3-C12; global demand forecast, OECD stocks
OPEC Monthly Oil Market Report	OPEC Sec.	Monthly	~2nd week of month	C3-C12; OPEC production data and demand view
OPEC+ Ministerial Meeting	OPEC+	2-4x per year	Variable	All months — most market-moving scheduled event
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET	Sentiment / positioning — all months
Canadian NEB export data	Canada NEB	Monthly	~6-8 wk lag	C3-C6; Canadian crude supply into US system
Texas RRC production data	Texas RRC	Monthly	~2 month lag	C3-C6; most accurate Permian / Eagle Ford data
North Dakota NDIC data	NDIC	Monthly	~2 month lag	C3-C6; Bakken production accuracy check
Kpler / Vortexa tanker data	Private	Daily/Weekly	Continuous	Real-time US export volumes and crude flows
IEA / EIA World Energy Outlook	IEA / EIA	Annual	Oct-Nov	C36+; long-run demand and transition scenarios

WTI crude oil is a market with a uniquely tractable physical anchor: Cushing inventory levels, measured weekly by the EIA, provide a direct, quantifiable connection between physical supply/demand balance and front-spread pricing. This makes the WTI prompt spread more mechanically predictable than most commodity spreads in the short run.

Beyond the prompt, WTI spreads are driven by the same global forces as any crude benchmark — OPEC+ policy, macro demand cycles, energy transition expectations — mediated through the specific structural features of the US market: shale production dynamics, pipeline infrastructure, and the export market that emerged post-2015. The edge in WTI spread trading comes from integrating the weekly Cushing data cycle with a rigorous view on the direction of US production, the pace of refinery demand, and the global pricing environment that determines export competitiveness.